



UNIVERSITY of
RWANDA

**COLLEGE OF SCIENCE AND
TECHNOLOGY
SCHOOL OF ICT**

DEPARTMENT OF COMPUTER AND SOFTWARE ENGINEERING

(NYARUGENGE Campus)

END OF TRIMESTER EXAMINATION -ACADEMIC YEAR 2024-2025

YEAR: 1 SEMESTER: II GROUP: Computer and Software Engineering

MODULE TITLE: DATABASE SYSTEMS

MODULE CODE: COE1262

DATE: 13/06/2025

TIME: 2 hours

MAXIMUM MARKS = 50

INSTRUCTIONS

1. This paper contains **FOUR (4)** questions.
2. **Answer THREE (3) Questions only:**
Question ONE (1) from Section "A" is Compulsory and Answer any TWO (2) from Section "B"
3. No written materials allowed.
4. No phones, no calculators.
5. Do not forget to write your Registration Number.
6. Write all your answers in the booklet provided
7. Do not write any answers on this questions paper.
8. Start each question in a NEW page.

SECTION : A

COMPULSORY QUESTION

Question 1

a) Multiple question /8pts

1. A software that enables user to define, create and maintain the database is:

- A. Oracle B. MySQL C. Database Management System D. Microsoft Office Access 2007.

2. Which of the following best describes a relation model in a database-----

- A. The data stored in the database. B. The overall structure and blueprint of the database. C. A software tool for managing data.

3. What is the term used to describe a single row or record in a relational database table?

- A. Cardinality B. Tuple C. Degree D. Domain

4. All UR colleges operate five schools. This relationship represents a -----

- A. one to one B. one to many C. many to one D. many to many

5. -----works on a single relation R and defines a relation that constrains only those tuples that satisfies a given condition.

- A. Cartesian B. Union C. Selection D. Difference

6. Which SQL keyword is used to delete the table -----

- A. DELETE B. TRUNCATE C. DROP D. REMOVE

7. Which of the following best describes a "foreign key"? -----

- A. A key that is unique within a table. B. A key that references the primary key of another table.
C. A key used for encryption in a database. D. A key that automatically generates sequential values.

8. Which of the following is a DCL (Data control Language) command?.....

- A. CREATE B. SELECT C. GRANT D. ALTER

- b) Explain with an example, three types of database anomalies that we can find in any normalized database /3pts
c) With help a diagram explain the transaction states /5pts
d) Explain the temporary update (dirty read) problem in concurrent transaction /2pts
e) What is the difference between homogeneous and heterogeneous distributed databases /2pts?

SECTION: B

OPTIONALY QUESTIONS: ANSWER ANY TWO

Question 2

Given the tables drinker(drinkerId, FirstName, LastName, Age, Sex, Address), Beverage(BeverageId, BeverageName, type, price), consumption(drinkerId, BeverageId, waitressId, BarId, date, quantity), Waitress(waitressId, FirstName, LastName, Age, Sex, Address), Bar(BarId, BarName, Location, Owner).

- I. perform the following queries using SQL command :/9pts
 - a) Create the Drinker table
 - b) Find the Beverage drunk by 'Louis'
 - c) Modify the drinker's FirstName called 'Peter' to 'Felicien'
 - d) Add the column 'Telephone' to the table drinker;
 - e) delete THE COLUMN's TELEPHONE'
 - f) Find the FirstName, LastName of all female drinkers whose age is under 18
 - g) Find the bar in which they serve the beverage called 'Guinness'.
 - h) Find all bars whose location is 'Kigali'
 - i) Find What is the average beverage price?

- II. Refer to hotel management system the following GUEST Table seems to be unnormalised, please use the concept of normalization to normalize it up to Third normal form. /6pts

HotelID	HotelName	GuestID	GuestName	GuestPhone	PaymentDate	PaymentAmount
H001	Grand Plaza	G001	John mugabo	787643210	5/1/2025	\$300
H002	Serena	G002	Jane sheja	7865432109	5/3/2025	\$500
H001	Grand Plaza	G003	Alice munyana	7854321098	5/2/2025	\$200
H003	Marriot	G004	Jonny Beza	786510987	5/4/2025	\$1,000

Question 3

In Rwanda there are farmers, from various regions, who sell their crops to different companies scattered all over the country. Many farmers may deliver the same crops and a region may host many farmers. Farmers are not limited to deliver their crops to the same companies, nor in the same region. Each delivery is recorded in a centralized database because the country wants to have statistics about how it is auto sufficient in producing basic crops.

- a) Draw an ERD diagram for that database. /6pts
- b) Map your ERD to the relational model /5pts
- c) From the above ERD identify weak and strong entity 4/pts

Question 4

- a) Concurrency in databases can be controlled using either lock-based protocols or time stamp-based protocols. How do these two protocols differ from each other? 5/pts
- b) As opposed to centralized databases there was a shift to distributed databases in today's application. In your understanding what are the three main raisons that make it possible. 5/pts
- c) List at least four advantage of distributed database application two for end users the remaining for the owners of the systems. 5/pts